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A study of Artificial Intelligence impacts on Human Resource Digitalization in Industry 4.0

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ABSTRACT

Artificial Intelligence (AI) has opened up tremendous opportunities in the workplace through robotics innovation, which envelops both AI and the Internet of Things (IoT). Precision, Efficiency, and Flexibility are considered the potential benefits of Industry 4.0. The implementation of Industry 4.0 requires a lot of changes, including the Human Resource (HR) function. In Industry 4.0, the HR capability is more critical and gives an upper hand to the organization. The HR capability should be more cautious and adaptable to adjust to the difficulties and requirements. We study the contributions of AI in HR digitalization and practices in Industry 4.0. 271 HR experts working in Information Technology (IT), Manufacturing, and administration are selected to participate in this review focusing on five AI applications in HR capability and three elements of HR readiness. The information collected was examined utilizing the Statistical Package for Social Sciences (SPSS) tool and Analysis of Moment Structures (AMOS). The results uncovered that hierarchical organization examination is a fundamental part of acquiring sustainable development. Adaptability and human asset capability are upheld by each of the five components of AI application areas of HR. Well-being and Safety improvement were viewed as vital components under the AI application in HR.

1. Introduction

In the era of Industry 4.0, the Human Resource (HR) function plays a critical role in bridging the gap between technology and human resources. Although technology is taking over most of the tasks traditionally performed by human resources, there is still a growing need for flexible HR functions to address the challenges of managing people. To achieve this flexibility, technology can help bring agility to the HR process. Agility, which is the ability to move quickly and smoothly, is not a new concept and has been adopted by major companies such as Google, Apple, Facebook, Amazon, and Microsoft. In the context of HR, agility means the ability to adapt and develop individuals and processes in response to rapid and unpredictable changes, to support people, key strategies, and organizational adaptability [1,2]. As an HR or Learning and Development (LandD) professional, being agile means being able to drive employee engagement and retention in alignment with the company's overall objectives. HR Agility is particularly suitable for volatile HR functions where standardization of functions is difficult [3,4].

In order for organizations to become agile, their workforce must focus on customer satisfaction and deliver value to them. However, since the HR function is not typically designed to provide direct incentives to the customer, it is often criticized for being slow to respond, which leads to dissatisfaction among employees. Therefore, organizations must encourage and empower HR to be more attentive and responsive to changing technologies and business needs in order to remain competitive and attract top talent. In an agile organization, HR continues to provide recruitment, development, performance management, and other HR functions, but using agile methodologies. There are three aspects of HR agility: the ability to quickly and efficiently identify issues that need to be addressed, the ability to reduce the time it takes to develop and implement a response, and the integration of analysis and design thinking to anticipate, plan, and target programs with the highest likelihood of success [2,5].

The rapid advancement of technology, especially the implementation of AI in HR, has brought about significant changes in HR processes and practices. As organizations increasingly move towards digitalizing their HR operations, it is crucial to understand the effects of AI on different aspects of HR such as employee productivity, health and safety, payroll processing, employee comfort, and real-time feedback [6]. Moreover, comprehending how these HR functions affect

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Table 1
Reliability and validity of the constructs.

Constructs	α	CR	AVE	MSV	ASV	HSI	EEC	EPM	APP	RTF	DHR	ONA	OD
Health and Safety Improvement	0.953	0.929	0.652	0.629	0.504	**0.807**							
Enhancing employee comfort	0.938	0.934	0.587	0.54	0.500	0.700	**0.766**						
Employee productivity measurement	0.928	0.933	0.581	0.51	0.462	0.731	0.681	**0.763**					
Automating payroll processing	0.978	0.966	0.778	0.537	0.332	0.607	0.571	0.532	**0.882**				
Real time feedback	0.901	0.895	0.523	0.509	0.380	0.684	0.592	0.533	0.555	**0.723**			
Digitization of HR	0.965	0.923	0.503	0.323	0.245	0.490	0.517	0.483	0.418	0.410	**0.709**		
Organizational Network Analysis	0.900	0.898	0.570	0.491	0.462	0.682	0.669	0.734	0.578	0.566	0.481	**0.754**	
Organizational design	0.945	0.915	0.730	0.645	0.544	0.790	0.766	0.716	0.618	0.672	0.548	0.737	**0.854**

Table 2
Demographic profile of respondents.

Demographic Variable	Category	No of Respondents	Percentage of the Respondents
Gender	Male	131	48.3
	Female	140	51.7
Age	21–30	89	32.8
	31–40	121	44.6
	41–50	42	15.5
	>50	29	10.7
Education	Master's degree	95	35.0
	Bachelor's degree	176	65.0
Industry Type	Manufacturing	80	29.5
	IT and ITES	129	47.5
	Service Sector	62	23.0

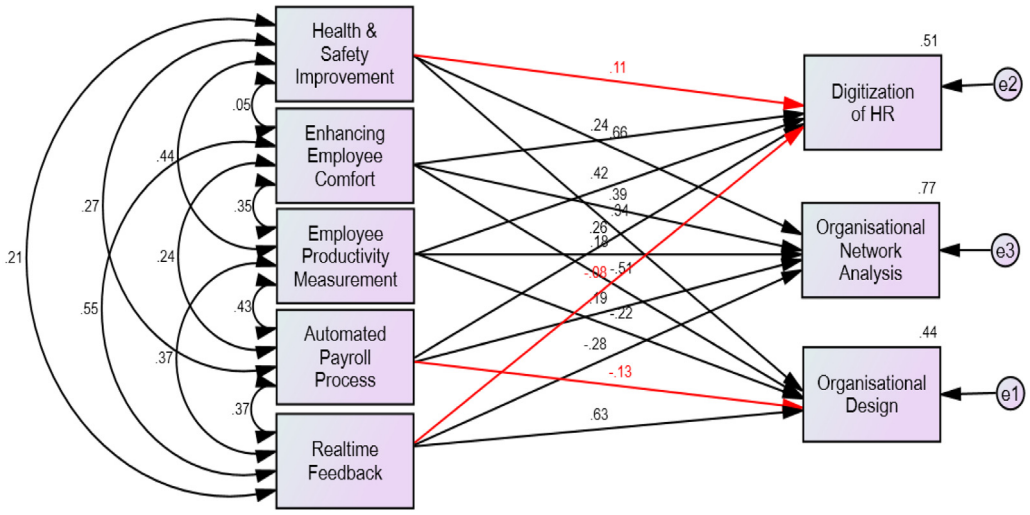

Fig. 2. Hypothesized Conceptual Model.

a test value of 874.98 and a degree of significance less than 0.0001. Therefore, the hypothesis that the correlation matrix is an identity matrix is rejected, meaning that the variables are not orthogonal. The significant value of less than 0.05 suggests that conducting a factor analysis on the dataset would be worthwhile.

4. Results, analysis and interpretation

In this section, the demographic profile of the respondent is discussed. Further, the results and interpretations are provided under this section.

4.1. Profile of the respondent

Table 2 depicts that 51.7% of the employees belong to the female category, whereas 48.3% of the data are received from the male respondents. Hence, females dominated the responses attained for this study. 44.6% of the respondent's age group are 31–40, and 32.8% are between 21 and 30. Hence a significant number of response was received from respondents of 31–40 years of age. Around 65% of the respondents hold a bachelor's degree, and the remaining respondents are have an education qualification of a master's degree. A majority of the respondents are pursuing a bachelor's degree. 47.5% of the respondents belong to IT and ITES organization, 29.5% of the respondents belong to the Manufacturing sector, and 23% of the respondents are employees of the service sector.

Structural Equation Modelling (SEM) technique was adopted to test the proposed conceptual model using Analysis of Moment Structures (AMOS 20) software [70]. It analyses the path relationship between the dimensions of AI application (i.e., Independent variables) and the dimensions of HR Agility (i.e., Dependent variable).

Fig. 2 portrays three causal relationships between AI Application to Digitization of HR, AI Application and Organisational Network Analysis, and AI Application and Organisational Design. The lines marked in red are statistically insignificant. From Table 4, the beta value and *P*-value belong to the above said variables' causal relationships are derived. All the five dimensions of AI in HR are showing highly significant value towards the dependent variable Organisational Network

Table 3
Fit indices of the conceptual model.

	CMIN/DF	RMSEA	CFI	IFI	GFI	AGFI	RMR	P
Model	1.563	0.043	0.997	0.997	0.993	0.868	0.006	0.154
Recommended Standard	<3.0	<0.08	>0.90	>0.90	>0.90	>0.90	<0.08	>0.05

Table 4
Results of the conceptual model.

Hypothesis	Path	Standardized Co-efficient	*P*-value	R^2
H1a	Employee Productivity Measurement → Digitization of HR	.422	***	0.508
H2a	Health and Safety Improvement → Digitization of HR	.109	.090	
H3a	Automated Payroll Process → Digitization of HR	.261	***	
H4a	Enhancing Employee Comfort → Digitization of HR	.238	***	
H5a	Real time Feedback → Digitization of HR	−.084	.247	
H4b	Real-time Feedback → Organisational Network Analysis	−.278	***	0.772
H5b	Enhancing Employee Comfort → Organisational Network Analysis	.386	***	
H6b	Health and Safety Improvement → Organisational Network Analysis	.660	***	
H1b	Employee Productivity Measurement → Organisational Network Analysis	.180	***	
H2b	Automated Payroll Process → Organisational Network Analysis	.194	***	
H6c	Enhancing Employee Comfort → Organisational Design	−.514	***	0.440
H5c	Health and Safety Improvement → Organisational Design	.339	***	
H4c	Employee Productivity Measurement → Organisational Design	−.222	.004	
H7	Automated Payroll Process → Organisational Design	−.129	.064	
H8	Real-time Feedback → Organisational Design	.630	***	

Analysis, and their beta values are −0.278, 0.386, 0.660 0.180, and 0.194 respectively. Out of five dimensions of AI in HR, only three dimensions have a significant influence on the Digitisation of HR, and their beta values are 0.422, 0.261, and 0.238 respectively. Out of five AI in HR dimension, one dimension named automated payroll system is not significant with the endogenous variable organizational design, the other dimensions are showing the significant influence on organizational design, and their beta values are −.514, 0.339, −0.222, and 0.630.

Table 3 displays the values of various goodness of fit indices. The values of the mentioned fit indices, normed chi-square is 1.563 P value = 0.154, RMR = 0.006, AGFI = 0.868, IFI = 0.997, GFI = 0.993, CFI = 0.997, RMSEA = 0.043. Except for AGFI, other values disclose the results regarding the fit of the proposed model within the threshold limits. The above values aptly fit the data in the proposed model.

Table 4 depicts the result of the three causal relationships between AI application areas and three dimensions of HR Agility as portrayed in the conceptual model as per Fig. 2. The five dimensions of AI application in HR, which are treated as exogenous variables in this study, are individually connected with each of the three endogenous variables, namely dimensions of HR Agility. Table 4 depicts that 77% of the variance in the Organisational Network Analysis is contributed by all the five dimensions of AI in HR. Out of that, Health and Safety Improvement are considered as a highly influencing dimension with a coefficient value of 0.660. The real-time feedback negatively contributes towards the variation in the organizational network analysis. On the other side, only three dimensions of AI applications are accounted for 51% of the variance in the Digitisation of HR. Employee productivity measurement tops the contribution with 0.422 as coefficient value. Four dimensions of AI in HR influence 44% of the variance in Organisational Design, and real-time feedback positively influence with a high coefficient value of 0.630. Surprisingly enhancing employee comfort negatively affects the organizational design with a negative coefficient value of −0.514.

5. Discussion

Employee health and well-being are a significant worry for managers as a solid specialist give different advantages to them, leading to expanded efficiency and income [71]. HR groups can use associated gadgets to screen and track worker well-being. Wearables can accumulate different information like food admission, strolling distance, and indispensable readings of representatives. Given the data accrued, HR staff can locate issues that are affecting well-being and causing medical problems and take suitable measures to keep away from them. HR needs to improve laborer well-being, and they can utilize AI to achieve that task. They can screen machines, hardware, and gas pipelines to protect their representatives [72]. For example, AI sensors can screen the crucial factor in gas pipelines to avoid any spillage because of the higher critical factor. The study identified that employee health and safety improvement is a highly influencing element in bringing agility in HR through ONA and organizational design.

A technology tracks the employees' eyes using sensors to identify its movement based on which HR personnel can distinguish factors like explicit work hours or foundation commotions that divert a worker. It can help HR people to assemble data on eye distraction while working on duty [73]. Suppose a worker feels snoozing during a couple of hours in the early evening, and their efficiency diminishes. HR can facilitate their employees to balance between their healthy living and working by arranging a life skill training program. This will ensure the employees actively concentrate on their work while they are on duty and promise higher productivity. This is the result of digitizing the HR process, which enhances agility. Here the results support the previous literature by [74–76].

It is sporadic to get honest feedback from employees in terms of official issues. Often, the HR department finds it hard to understand employees' real feelings and emotions and come out with a lot of strategies, but none of them are giving fruitful results [77]. This can be sorted out by incorporating AI applications. AI gadgets can help the HR personnel to understand the genuine emotions of their workforce while gathering criticism. Cameras can catch pictures of a representative after a gathering to accomplish constant criticism. The images can be sent absurd to the workers where computerized vision can identify feelings of the representative and send cautions to HRM's if a worker is not sincerely glad. Designing an effective organization to adopt a dynamic environment can be possible with the correct feedback [78]. AI made this possible ensures agility in the HR function. The result adhere to previous studies proposed by [3,76].

With the assistance of AI, AI can recognize designs characteristic of sadness and other psychological maladjustments in workers. Automated cameras can click pictures of workers at specific interval during the whole day. Computerized vision can remove personal conduct standards data from those pictures and contrast them, and those of discouraged individuals decide if a worker is experiencing uneasiness or desolation. Assuming computerized vision tracks down that a worker feels discouraged, it can impart signs to AI gadgets that can alarm HR.

HR staff can coordinate guiding meetings for that worker to improve his consolation at work. This may negatively impact the organizational design as identified by the results from existing study, and it is in contrast with the study by [79–81].

AI sensors to follow truancy can be executed for all positions, yet not to screen exact work hours. For example, administrative center positions expect workers to sit on their work areas for being profitable, and thus sensors can be executed for such jobs. Notwithstanding, field occupations do not need that, and AI sensors cannot be carried out to follow work hours for these sorts of professions. This is following the result proposed from the study by [74].

6. Conclusion, limitations and scope for future study

Carrying out AI in HRM gives numerous benefits to the HR department and employees. However, those advantages accompany a few network safety dangers and lawful concerns. Assembling more worker information implies expanded protection concerns, and more gadgets accompany more noteworthy prospects of network safety assaults. Before executing AI for HR the board, organizations need to ensure that their workers' information is not undermined. Associations likewise need to fabricate information-driven security to screen information itself and not simply organization to limit network protection dangers.

The study has addressed the application of AI concepts in various possible areas of HRM. These areas may not be there in the regular stream of activities. Still, it is trying to impose the importance of addressing the same—the dimensions measured under that showcase human aspects enhancement with the help of AI. The result revealed the way these aspects influence the agile capability of HRM. Digitization of HR and ONA is closed related technological implication in HR, enabling the iterative process of function. Both of the above aspects need an excellent organizational design to support the implementation and progress. Thus this study paved a new way by linking two elements prevailing today's Industry 4.0 era.

There are not many businesses using AI in their HRM or creating AI based HR software because it is still a very new and low-use field, particularly in India. As a result, it is challenging to compile a thorough study because the majority of businesses only use AI to a limited degree in the HR process. Even though AI as a subject has received extensive research, the ability to examine the true effectiveness and ramifications that AI involves is limited in the absence of a sufficient number of organizations who employ AI in their HR practices. The number of interviews may be larger to make this study more relevant. The responses of the interviewees, however, may be compared and contrasted.

As this study has demonstrated, the use of AI in recruitment is still a relatively new topic. More AI-related research should be conducted in the future to get a better picture of the subject. Although empirical findings from several organizations were used in this study, when more information about AI becomes available, an organization-specific study could be conducted. Organizations that do not currently use AI but intend to do so in the future could be included in the study to gain a broader perspective on the subject. Despite the potential benefits of AI in HR, there are also challenges and concerns that need to be addressed. One of the main concerns is the potential for bias in AI algorithms. If AI algorithms are trained on biased data, they may perpetuate and even amplify biases in the HR practices. Another concern is the potential for job displacement due to automation. As AI takes over more HR functions, there is a risk that some HR professionals may lose their jobs.

Furthermore, using a quantitative approach, it could be investigated how AI based HR decisions have impacted company success and turnover in numerical terms. Because there are trust issues with AI, employees' perspectives and experiences with AI-based HR practices could be studied to gain more perspectives on this topic.

List of Abbreviations

1. Artificial Intelligence (AI)
2. Human Resource (HR)
3. Internet of Things (IoT)
4. Statistical Package for Social Sciences (SPSS)
5. Structural Equation Modelling (SEM)
6. Analysis of Moment Structures (AMOS)
7. Information Technology (IT)
8. Human Resource Management Practices (HRMP).
9. Confirmatory factor analysis (CFA)
10. Composite reliability (CR)
11. Average variance extracted (AVE)
12. Maximum Shared Value (MSV)
13. Average Shared Value (ASV)
14. Kaiser-Meyer-Olkin (KMO)

Declaration of competing interest

The authors have no conflicts of interest to declare. All co-authors have seen and agree with the contents of the manuscript and there is no financial interest to report.

Data availability

Data will be made available on request.

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